1-) Project Name

DiaRisk: Diabetes Risk Screening with Machine Learning for SDG 3

2-) Overview

This project aims to build an AI-based tool that estimates a person's risk of diabetes or prediabetes using only simple, self-reported health and lifestyle information, without a blood test. Using the "Diabetes Health Indicators" dataset, we analyse how factors such as BMI, blood pressure, cholesterol, physical activity, age and income are related to diabetes. The project is aligned with the following United Nations Sustainable Development Goals:

- **SDG 3 (Good Health and Well-being), Target 3.4:** This target aims to reduce premature deaths from non-communicable diseases such as diabetes through prevention and early treatment.
- **SDG 10 (Reduced Inequalities):** This goal aims to reduce inequalities within countries. Diabetes risk is higher among low-income groups, and a free screening tool that requires no blood test can reach people with limited access to healthcare.

Problem Statement: Diabetes is a growing health problem in Türkiye. According to the IDF Diabetes Atlas, about 16.5% of adults in Türkiye have diabetes and almost half of them (45.5%) do not know it. Diabetes often develops silently and is noticed only after serious complications appear. Prediabetes can often be reversed with lifestyle changes, but only if the person is aware of it. Since diagnosis requires a blood test, many people without symptoms or with limited access to healthcare are never tested.

Analytical Approach: To address this problem, the project starts with a careful analysis of the data before any model is built. We aim to find which health and lifestyle factors are most strongly related to diabetes and how the risk changes across BMI, age and income groups. For example, the analysis investigates whether the risk increases steadily with BMI and age, and whether people with lower income carry a higher risk. It also examines data quality problems such as duplicates, outliers, skewed variables and class imbalance, which directly affect how the model should be trained and evaluated.

Impact Potential: Our tool will let a user answer a short questionnaire and receive a risk level (low, medium or high) together with the factors that increase their risk most. An LLM will then turn this result into a personal lifestyle plan in plain language. By encouraging people at risk to change their habits and see a doctor earlier, the tool can support early diagnosis and contribute directly to SDG 3.

3-) Background

Type 2 diabetes is strongly linked to risk factors that can be measured without a laboratory test, such as obesity, high blood pressure, physical inactivity and age. Questionnaire-based risk scores have long been used to decide who should be tested, and machine learning can improve them by learning more complex relationships from large datasets. The "Diabetes Health Indicators" dataset, a cleaned version of the US CDC's 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey, contains 253,680 responses and 21 health and lifestyle features.

Since a risk score alone is often not enough to change behaviour, our project combines a machine learning risk model with an LLM that explains the result in a clear and personal way. Analysing the data is the first step and provides a clean, well-understood foundation for both parts.

4) Key Objectives / Business Objectives

a) Research Questions:

1. Which self-reported health and lifestyle factors are most strongly related to diabetes and prediabetes?
2. How do BMI, age and income affect the risk of diabetes?
3. How does the strong class imbalance in the data affect the way the model should be built and evaluated?
4. Can simple questions without a blood test give a reliable estimate of diabetes risk?
5. How can the most important risk factors be explained to users in an understandable way?

b) Key Steps:

1. Clean and preprocess the dataset (missing values, duplicates, outliers, data types).
2. Perform exploratory data analysis with descriptive statistics and visualisations.
3. Analyse the relationship of each feature with diabetes using correlation analysis.
4. Train and compare machine learning models to predict diabetes risk.
5. Integrate an LLM that explains the result and creates a personal lifestyle plan.
6. Test the tool with different user profiles and evaluate its results.

5) Methods and Workflow

a) Datasets:

- **Source:** "Diabetes Health Indicators Dataset" (Kaggle), based on the CDC BRFSS 2015 survey, with 253,680 rows and 22 columns.
- **Scope:** The target variable Diabetes_012 has three classes (0 = no diabetes, 1 = prediabetes, 2 = diabetes). The 21 features include health conditions (high blood pressure, high cholesterol, BMI), lifestyle habits (smoking, physical activity, diet) and demographic information (age, education, income).

b) Data Cleaning and Preprocessing:

1. **Missing Values and Duplicates:** The dataset has no missing values. 23,899 duplicate rows were removed.
2. **Outlier Removal:** 584 rows with an unrealistic BMI value above 70 were removed. The final dataset has 229,197 rows.
3. **Data Transformation:** All columns were converted from float to integer, and a binary target (0 = no diabetes, 1 = prediabetes or diabetes) was created for the risk analysis.
4. **Skewness Correction and Scaling:** Preparing the variables for balanced, multi-variable modelling:
    - Applying a log transformation to the right-skewed BMI variable, which reduces its skewness from 1.19 to 0.42 (also applied to the health-day variables MentHlth and PhysHlth).
    - Utilizing Standard Scaler for continuous variables (BMI, MentHlth, PhysHlth).
    - Utilizing Min-Max Scaler for ordinal variables (general health, age group, education, income); binary 0/1 variables need no scaling.
    - The scalers will be fitted only on the training data after the train/test split in Assignment 2, to avoid data leakage.

c) Modelling (EDA Phase):

1. **Descriptive Statistics and Class Distribution:** The classes are highly imbalanced: 82.7% no diabetes, 2.0% prediabetes and 15.3% diabetes. BMI and the health-day variables are right-skewed.
2. **Risk Factor Analysis:** The diabetes/prediabetes rate rises from 7.8% at normal weight to 33.5% at BMI 35 or higher, and from 1.8% at ages 18–24 to 25.8% at ages 70–74. It falls from 27.5% in the lowest income group to 11.2% in the highest (Figure 1).
3. **Correlation Analysis:** A Spearman correlation heatmap was created, since most variables are binary or ordinal (Figure 2). General health (0.28), high blood pressure (0.26) and BMI (0.23) are most related to diabetes.
4. **Multicollinearity Control:** No feature pair has a correlation above 0.85 (the highest is 0.44), so no feature needs to be removed.

How the Dataset Supports the Project Goals: The EDA shows that simple, self-reported answers that require no blood test carry a clear and medically meaningful signal about diabetes risk, which confirms that a no-lab screening tool is feasible. The strongest factors (general health, blood pressure, BMI and physical activity) are also easy for users to understand and partly change, which makes them suitable for the tool's explanations and lifestyle plans.

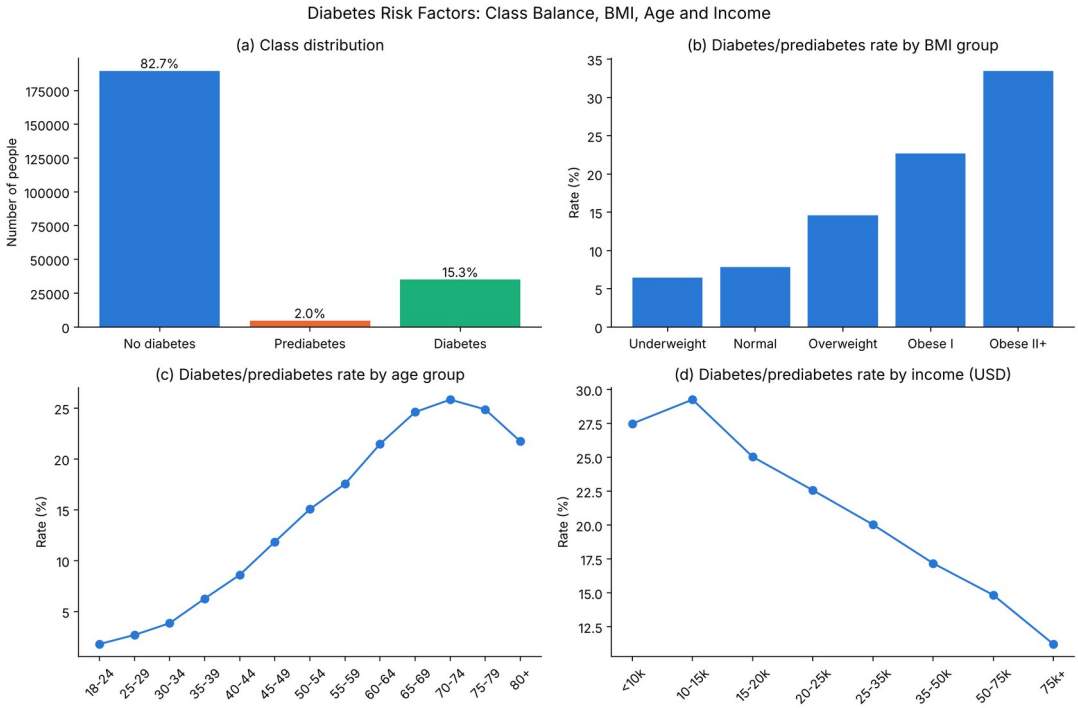


Figure 1. Class distribution and diabetes/prediabetes rate by BMI, age and income

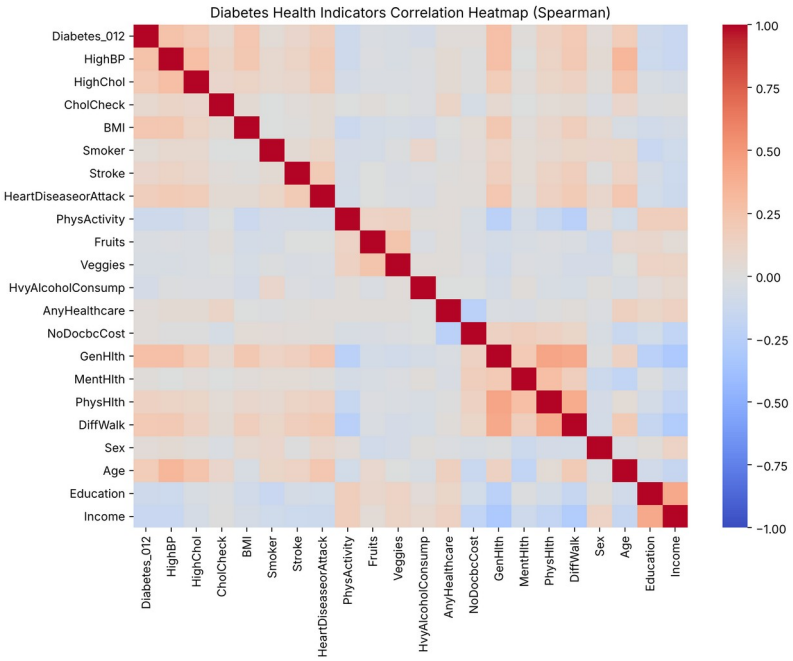


Figure 2. Spearman correlation heatmap of all variables

d) Deliverables:

1. A Python notebook with the data cleaning, preprocessing and EDA steps.
2. Visual reports including risk factor charts and a correlation heatmap.
3. A cleaned dataset ready for machine learning models in the next assignment.

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